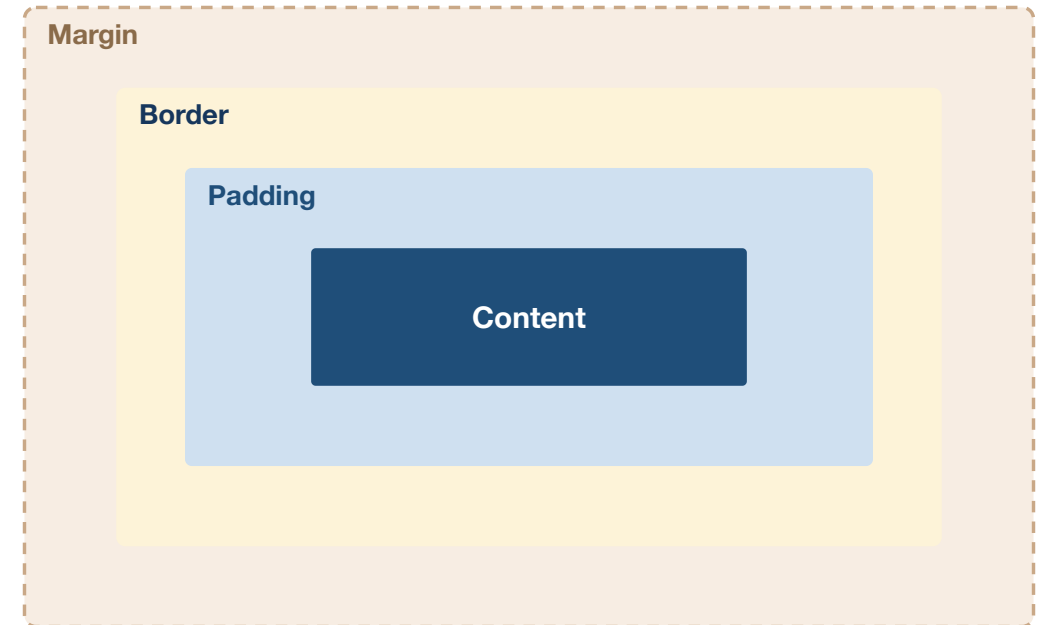


The Box Model & the Cascade

Full Stack: Frontend, Backend, Database & Deployment · Class 13

What Is the Box Model?

- Every HTML element renders as a rectangular box
- Four layers, inside out: content, padding, border, margin
- DevTools' Elements panel shows these layers directly

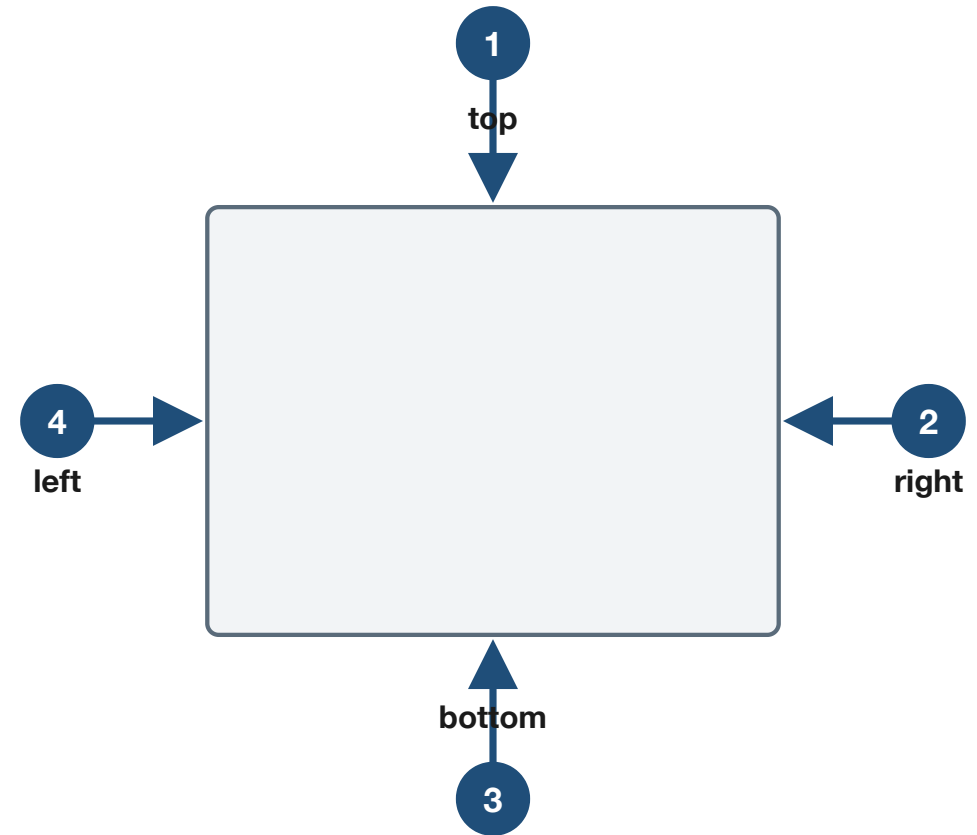


The Four Layers

- **Content** — the text/media itself
- **Padding** — space between content and border, painted with the background color
- **Border** — a visible line drawn around the padding
- **Margin** — transparent space outside the border, separating it from neighbors

Padding & Margin Shorthand

- One value — all four sides
- Two values — top/bottom, then left/right
- Three values — top, then left/right, then bottom
- Four values — top, right, bottom, left



Centering with margin: auto

- Needs a fixed `width` — auto margins split whatever space is left over
- `margin: 0 auto;` centers a block box horizontally
- `auto` doesn't center vertically — there's no leftover space to split the same way top/bottom

Margin Collapsing

- Adjacent top/bottom margins between stacked elements collapse into one gap
- The gap is the larger of the two margins, not their sum
- Padding never collapses; left/right margins never collapse



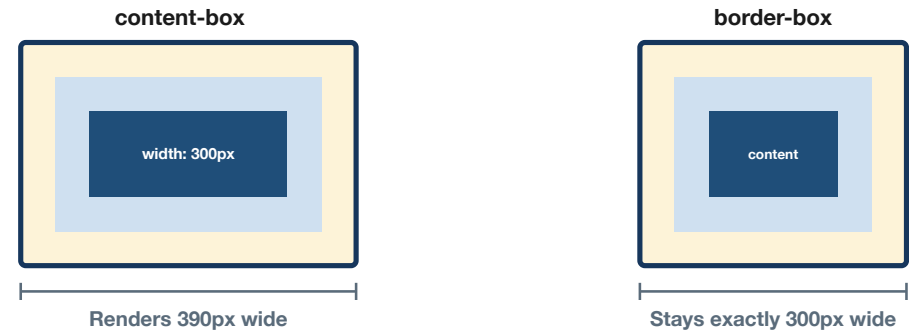
Actual gap: 30px (the larger one) — not 50px

content-box (the Default)

- `width` sets the size of the content only
- Padding and border are added on top of that width
- A 300px box with 40px padding and a 5px border renders 390px wide

border-box

- `box-sizing: border-box;` changes what `width` measures
- `width` now includes content, padding, and border together
- The box stays exactly the width you set it to



Same width, padding, and border on both boxes

The Universal Selector

- `*` matches every single element on the page
- Includes elements you never wrote a rule for — even `<html>` and `<body>`
- One rule instead of writing the same rule for every tag by hand

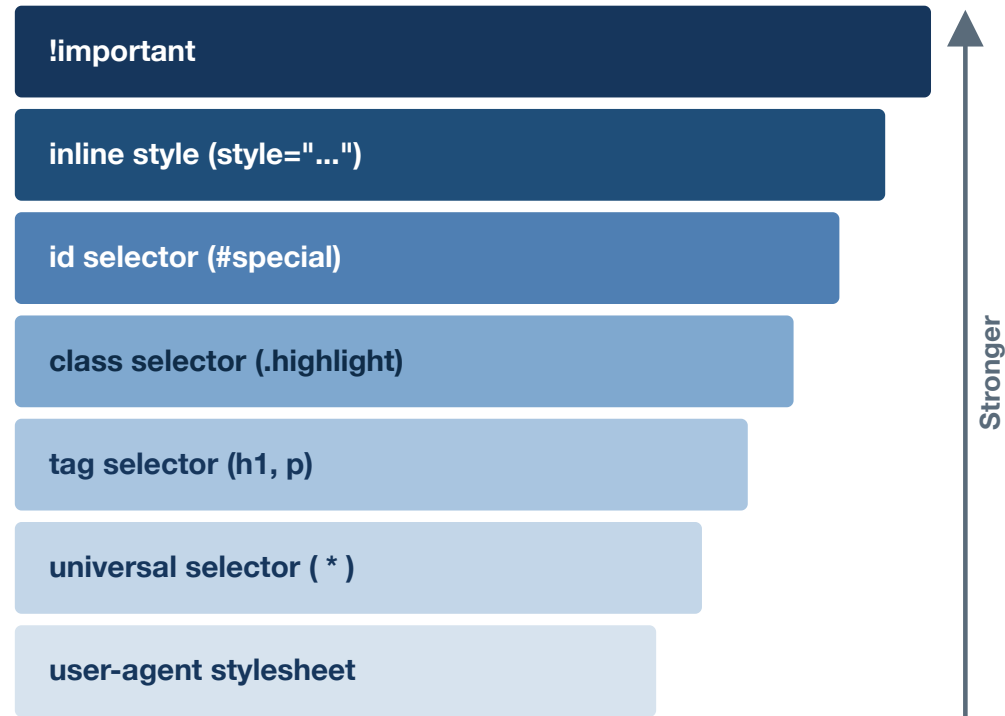
The User-Agent Stylesheet

- Every browser ships its own default CSS, applied before any CSS you write
- Why an unstyled `<h1>` is already bold/large, and links are already blue and underlined
- Sits at the very bottom of the cascade — any rule you write beats it

Specificity — Who Wins?

- Multiple rules can target the same element at once
- The browser has to pick one winner per property
- That winner is decided by **specificity** — how "targeted" a selector is

The Specificity Ladder



px — Absolute

- A fixed, literal size in pixels
- Never changes when the window or parent element is resized
- The most predictable unit, but the least flexible

% — Relative to the Parent

- Sized as a percentage of the containing (parent) element
- Resize the parent, and the element resizes with it
- If no ancestor has a set width, it traces back to the browser window itself

`vw` & `vh` — Relative to the Viewport

- `vw` — 1% of the viewport's width; `vh` — 1% of its height
- Measured from the browser window, not any parent element
- The only one of the four always tied to the window itself

